AI FAQ ASSISTANT - PROJECT REPORT

Phase 4: Project Planning Phase

Technology Stack

• Runtime: Node.js
• Framework: Express.js
• Database: MongoDB with Mongoose
• Security: JWT + bcrypt
• AI: Google Gemini
• Search: Keyword-based search
• Testing: Postman/ThunderClient

Project Structure

Directory structure organized by responsibility:
src/
■■■ config/db.js
■■■ controllers/
■■■ middleware/
■■■ models/
■■■ routes/
■■■ services/
■■■ utils/
■■■ app.js
■■■ server.js

Project Timeline

• Phase 1: Project initialization and setup
• Phase 2: Database schema and models
• Phase 3: Authentication and authorization
• Phase 4: FAQ CRUD operations
• Phase 5: AI integration
• Phase 6: Testing and bug fixes
• Phase 7: Documentation
• Phase 8: Demonstration and presentation

Environment Configuration

• PORT: API server port
• MONGO_URI: Database connection string

• JWT_SECRET: Token signing secret
• GEMINI_API_KEY: Google Gemini API key

Development Tools

• VS Code for code editing
• Postman/ThunderClient for API testing
• Git for version control
• npm for dependency management

Generated on: 29-09-2026 18:58